

$$p(t) = 4t^3$$

describes precisely those fluctuations.

This connects directly to the issue we discussed earlier about the diagrams: the **actual X_i 's in a run are irregular**, while the statistical law governing their irregularity is

$$t_i = \frac{X_i}{X_{i-1}} \sim \text{Beta}(N_{\text{live}}, 1).$$

The deterministic-looking approximation

$$X_i \simeq e^{-i/N_{\text{live}}}$$

is only the typical log-volume trajectory through that stochastic process.

One particularly compact way to remember the whole derivation is

$$\begin{array}{l} N \text{ live points} \\ \Downarrow \\ u_1, \dots, u_N \stackrel{\text{iid}}{\sim} U(0, 1) \\ \Downarrow \\ t = \max(u_1, \dots, u_N) \\ \Downarrow \\ P(t < T) = T^N \\ \Downarrow \\ p(t) = Nt^{N-1} \\ \Downarrow \\ t \sim \text{Beta}(N, 1). \end{array}$$

And the perhaps even more useful equivalent statement is

$$-\log t \sim \text{Exp}(N_{\text{live}}).$$

That form makes the approximately constant rate of progress in $\log X$ especially transparent.



Today 9:48 AM

I am still unclear at the very beginning, where you conclude that $X(\theta)$ is uniformly distributed between zero and 1. First, what is $X(\theta)$, given that X is a function of λ ? Is it $X(\lambda(\theta))$? Then give me more detail on why a uniform distribution follows.

Yes. The notation $X(\theta)$ is shorthand, and it is worth making it explicit because otherwise the argument looks mysterious.

Start with the actual definition:

$$X(\lambda) = \int_{L(\theta) > \lambda} \pi(\theta) d\theta.$$

So X is fundamentally a function of a **likelihood threshold** λ , not directly of θ .

Given a particular point θ , however, that point has a likelihood value

$$\lambda = L(\theta).$$

We can therefore associate with the point the prior volume enclosed by its likelihood contour:

$$\boxed{X(\theta) \equiv X(L(\theta)).}$$

So yes: what I previously called $X(\theta)$ is precisely the composition

$$\theta \longrightarrow L(\theta) \longrightarrow X(L(\theta)).$$

That is the first important clarification.

1. What does $X(\theta)$ mean geometrically?

Suppose we have a two-dimensional parameter space. Pick a point θ_* . It has likelihood

$$L_* = L(\theta_*).$$

Draw the iso-likelihood contour through that point. Then

$$X(\theta_*) = \int_{L(\theta) > L_*} \pi(\theta) d\theta$$

is the total **prior probability lying inside the higher-likelihood region**.

So if θ_* is way out in a low-likelihood region,

$$X(\theta_*) \approx 1.$$

If it is very close to the likelihood maximum,

$$X(\theta_*) \approx 0.$$

Thus $X(\theta)$ assigns every point a number between 0 and 1 according to how much prior probability has likelihood larger than that point's likelihood.

This is why the contours in your figures can equally well be labeled by L or by X .

2. Now draw Θ randomly from the prior

Here it is useful to distinguish a random parameter point Θ from a particular value θ :

$$\Theta \sim \pi(\theta).$$

Its likelihood

$$Y = L(\Theta)$$

is now itself a random variable.

The distribution of Y is whatever distribution of likelihood values is induced by drawing points from the prior.

Define its cumulative distribution function

$$F_Y(\lambda) = P(Y \leq \lambda).$$

But

$$Y = L(\Theta),$$

so

$$F_Y(\lambda) = P_\pi(L(\Theta) \leq \lambda).$$

Meanwhile our nested-sampling prior volume is

$$X(\lambda) = P_\pi(L(\Theta) > \lambda).$$

Assuming a continuous likelihood distribution so that equality has zero probability,

$$X(\lambda) = 1 - F_Y(\lambda).$$

So $X(\lambda)$ is simply the **survival function of the likelihood when θ is sampled from the prior**.

That makes the uniformity result a version of the ordinary probability-integral transform.

3. Direct derivation of the uniform distribution

Define

$$U = X(L(\Theta)).$$

I will now show explicitly that

$$U \sim \text{Uniform}(0, 1).$$

Pick some number x between 0 and 1.

Because $X(\lambda)$ decreases monotonically as λ increases, there is a likelihood threshold λ_x such that

$$X(\lambda_x) = x.$$

In other words,

$$P_\pi(L > \lambda_x) = x.$$

Now ask:

$$P(U < x) = P(X(L(\Theta)) < x).$$

Since $X(L)$ gets **smaller** as L gets larger,

$$X(L(\Theta)) < x$$

means precisely

$$L(\Theta) > \lambda_x.$$

Therefore

$$P(U < x) = P(L(\Theta) > \lambda_x).$$

But λ_x was defined so that

$$P(L(\Theta) > \lambda_x) = x.$$

Hence

$$\boxed{P(U < x) = x.}$$

But that is exactly the CDF of a uniform random variable on $[0, 1]$:

$$F_U(x) = x.$$

Therefore

$$\boxed{X(L(\Theta)) \sim U(0, 1).}$$

That is the fundamental nested-sampling result.

4. A geometric way to see it

This may be even more intuitive.

Imagine dividing the prior into nested likelihood regions with prior volumes

$$X = 0.1, \quad 0.2, \quad 0.3, \dots, 1.$$

The region satisfying

$$X(\theta) < 0.2$$

contains, by construction, exactly 20% of the prior probability.

Therefore, if you draw a random point from the prior,

$$P(X(\Theta) < 0.2) = 0.2.$$

Similarly,

$$P(X(\Theta) < 0.7) = 0.7.$$

And in general,

$$P(X(\Theta) < x) = x.$$

There is no dependence here on whether the likelihood contours are circular, elliptical, multimodal, or horribly shaped. The transformation X has been **defined in terms of enclosed prior probability**, so it automatically turns prior draws into uniform random variables.

That is the same logic behind turning an arbitrary random variable Y into

$$F_Y(Y),$$

which is uniform.

5. A simple one-dimensional example

Suppose

$$\theta \sim U(0, 1)$$

and take a monotonically increasing likelihood

$$L(\theta) = \theta.$$

For a likelihood threshold λ ,

$$X(\lambda) = P(L(\theta) > \lambda) = P(\theta > \lambda).$$

Since θ is uniform,

$$X(\lambda) = 1 - \lambda.$$

For a particular point θ ,

$$L(\theta) = \theta,$$

so

$$X(\theta) = X(L(\theta)) = 1 - \theta.$$

But if

$$\theta \sim U(0, 1),$$

then clearly

$$1 - \theta \sim U(0, 1).$$

So in this elementary example the result is completely transparent.

6. A less trivial example: Gaussian likelihood

Suppose the prior is uniform over some large interval and

$$L(\theta) = e^{-\theta^2/2\sigma^2}.$$

This is not one-to-one in θ : both $+\theta$ and $-\theta$ have the same likelihood.

That doesn't matter.

For a particular likelihood level λ , the condition

$$L(\theta) > \lambda$$

defines an interval around the origin,

$$|\theta| < r(\lambda).$$

Then $X(\lambda)$ is simply the fraction of prior probability inside that interval.

A point at radius r therefore gets assigned

$$X(\theta) = \text{prior probability inside } |\theta'| < |\theta|.$$

If you randomly draw θ from the prior, the probability that this enclosed volume is less than x is exactly x .

Again,

$$X(\Theta) \sim U(0, 1).$$

So the argument does **not** require $L(\theta)$ itself to be monotonic in the original parameters.

7. This explains the constrained version used in nested sampling

Now suppose nested sampling has already reached prior volume

$$X_{i-1}.$$

The live points are drawn from the constrained prior

$$L > L_{i-1}.$$

In X -coordinates, that constraint is simply

$$0 < X < X_{i-1}.$$

Because X was uniform under the original prior, conditioning on

$$X < X_{i-1}$$

gives

$$X_j \mid X < X_{i-1} \sim U(0, X_{i-1}).$$

Then rescale:

$$u_j = \frac{X_j}{X_{i-1}}.$$

It follows immediately that

$$u_j \sim U(0, 1).$$

And *that* is the step that leads to

$$t_i = \max(u_1, \dots, u_{N_{\text{live}}})$$

and hence

$$t_i \sim \text{Beta}(N_{\text{live}}, 1).$$

So the whole chain is

$$\Theta \sim \pi \Rightarrow Y = L(\Theta) \Rightarrow X(Y) = P_\pi(L > Y) \Rightarrow X(Y) \sim U(0, 1)$$

The critical conceptual point is that X is **constructed to be a probability-content coordinate**. Its uniformity under prior sampling is therefore not a special property of the likelihood; it follows from the definition of X .

One qualification: this clean statement assumes that the distribution of likelihood values under the prior is continuous, so there are no finite-prior-mass plateaus at exactly the same likelihood. If there are discrete likelihood levels or plateaus, ties require a slight generalization.



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